MF 281: Probability Theory

Week 4: Independence

When knowing one thing tells you nothing about another

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Lecture notes for students of MF 281 Probability Theory.

Week 4: Independence

Learning Objectives

By the end of this week you should be able to:

- define independence by the product rule, and say why that form is preferred;
- show that disjoint events with positive probability are never independent;
- prove that independence survives complementation;
- distinguish pairwise from mutual independence, and give a counterexample;
- define conditional independence, and show it neither implies nor is implied by independence;
- compute with independent trials, including "at least one" problems;
- find the reliability of series, parallel and mixed systems.

4.1 The definition

Last week, conditioning changed what we believed. Independence is the case where it does not.

Definition 4.1 (Independent events)**.** A and B are independent if

$$P(A \cap B) = P(A)\,P(B).$$

When $P(B) > 0$ this is equivalent to $P(A \mid B) = P(A)$, which is the intuitive form: learning B leaves A exactly where it was. The product form is taken as the definition because it is symmetric in A and B, and because it still makes sense when $P(B) = 0$, where the conditional is undefined.

Independence is a statement about *numbers*, not about physical separation. Two events can be causally unrelated and still dependent in a given population, and two events can be physically linked yet satisfy the product rule by coincidence. Always check the arithmetic rather than the story.

Example 1: independent or not

A fair die is rolled. Let $A = \{2, 4, 6\}$ (even), $B = \{4, 5, 6\}$ (at least 4) and $C = \{1, 2, 3, 4\}$.

Solution

$P(A) = 1/2$, $P(B) = 1/2$ and $P(A \cap B) = P(\{4, 6\}) = 1/3$. Since $1/3 \neq 1/4$, A and B are **dependent**: knowing the roll is at least 4 makes even more likely (2/3 rather than 1/2).
Now $P(C) = 2/3$ and $P(A \cap C) = P(\{2, 4\}) = 1/3 = (1/2)(2/3)$, so A and C **are** independent.
Nothing about the physical set-up distinguishes these two cases: it is purely arithmetic.

4.2 Independence is not disjointness

This is the error that costs most marks in examinations.

Proposition 4.1. *If $P(A) > 0$, $P(B) > 0$ and $A \cap B = \varnothing$, then A and B are* dependent.

Proof. $P(A \cap B) = P(\varnothing) = 0$, while $P(A)P(B) > 0$ as a product of two positive numbers. The two differ, so the product rule fails. □

Intuitively: disjoint events are maximally informative about each other, because if one happens the other certainly did not. That is the opposite of telling you nothing.

Proposition 4.2 (Complements)**.** *If A and B are independent then so are A and B^c.*

Proof. $A = (A \cap B) \cup (A \cap B^c)$ disjointly, so

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)\big(1 - P(B)\big) = P(A)P(B^c). \quad \square$$

Applying it twice shows A^c and B^c are independent too, which is what justifies the standard move

$$P(\text{none of them occur}) = \prod_i \big(1 - P(A_i)\big)$$

for independent events.

4.3 Many events: pairwise is not enough

Definition 4.2 (Mutual independence)**.** Events $A_1, \ldots, A_n$ are mutually independent if for *every* subset $S \subseteq \{1, \ldots, n\}$,

$$P\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} P(A_i).$$

Checking all pairs is not sufficient, which surprises almost everyone.

Example 2: pairwise independent, jointly impossible

Toss a fair coin twice. Let A be "first toss is heads", B "second toss is heads" and C "exactly one head".

Solution

Each has probability 1/2. For the pairs:

$$P(A \cap B) = P(\{HH\}) = \tfrac{1}{4} = P(A)P(B), \quad P(A \cap C) = P(\{HT\}) = \tfrac{1}{4}, \quad P(B \cap C) = P(\{TH\}) = \tfrac{1}{4},$$

so all three pairs are independent. But

$$P(A \cap B \cap C) = P(\varnothing) = 0 \neq \tfrac{1}{8} = P(A)P(B)P(C),$$

because if both tosses are heads there is not exactly one head. The three events are pairwise independent and not mutually independent: any two of them together determine the third.

The moral for applications: "the components fail independently in pairs" is not enough to multiply three probabilities together. Mutual independence is an assumption about every combination, and it should be stated as such.

4.4 Conditional independence

Definition 4.3. A and B are conditionally independent given C, with $P(C) > 0$, if

$$P(A \cap B \mid C) = P(A \mid C)\, P(B \mid C).$$

Neither kind of independence implies the other.

Example 3: conditioning creates dependence, and destroys it

(a) Independent, but dependent given C. Toss two fair coins. A and B, the two heads, are independent. Let C be "exactly one head". Given C,

$$P(A \mid C) = P(B \mid C) = \tfrac{1}{2}, \qquad P(A \cap B \mid C) = 0 \neq \tfrac{1}{4}.$$

Knowing exactly one coin is a head, the two coins are now perfectly informative about each other.

(b) Dependent, but independent given C. Two machines draw from the same power supply. Their failures are associated, because a voltage dip takes both down. Given the state of the supply, they fail independently; the dependence was entirely explained by the common cause.

Why this matters

Part (b) is the engineering case that matters. Components in one cabinet are rarely independent, because they share a cause. Conditioning on that cause is often what makes a reliability calculation legitimate.

4.5 Independent trials

Repeating an experiment under identical conditions, with outcomes that do not influence one another, gives *independent trials*. For n trials each succeeding with probability p, a particular sequence with k successes has probability $p^k(1-p)^{n-k}$, and there are $\binom{n}{k}$ such sequences.

Example 4: four bolts

Twenty percent of bolts from a machine are defective, independently.

Solution

Exactly two defective in four:

$$\binom{4}{2}(0.2)^2(0.8)^2 = 6(0.04)(0.64) = 0.1536.$$

At least one defective, by the complement:

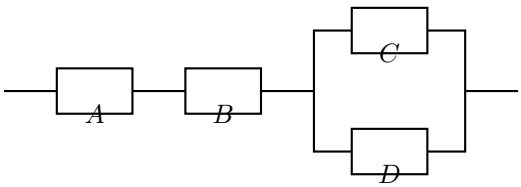
$$1-(0.8)^4 = 1-0.4096 = 0.5904.$$

None defective is 0.4096, and the two sum to one, as they must. That complement step is the single most useful consequence of independence in this course.

4.6 Reliability

Treat a system as components that work or fail independently.

- **Series:** the system works only if every component does, so $P(\text{works}) = \prod_i p_i$. Adding components makes it worse.
- **Parallel:** it works unless all fail, so $P(\text{works}) = 1 - \prod_i (1-p_i)$. Adding components makes it better.



A and B in series, with C and D in parallel

Figure 4.1: A mixed system. Reduce the parallel block to a single equivalent component, then multiply along the series.

Example 5: a mixed system

The system works if A and B both work and at least one of C, D works. The components are independent with $P(A) = P(B) = 0.9$ and $P(C) = P(D) = 0.8$.

Solution

Collapse the parallel block first:

$$P(C \cup D \text{ works}) = 1 - (1 - 0.8)(1 - 0.8) = 1 - 0.04 = 0.96.$$

Then the series:

$$P(\text{system works}) = 0.9 \times 0.9 \times 0.96 = 0.7776.$$

Now a conditional question

Given that the system works, what is the probability that C has failed?

$$P(C^c \cap \text{works}) = \underbrace{0.9 \times 0.9}_{A,B} \times \underbrace{0.2}_{C \text{ fails}} \times \underbrace{0.8}_{D \text{ works}} = 0.1296,$$

$$P(C^c \mid \text{works}) = \frac{0.1296}{0.7776} = 0.1667.$$

So one working system in six is running on D alone, which is exactly the number a maintenance planner needs: the system looks healthy while its redundancy is already spent.

Redundancy buys reliability only if the failures really are independent. Two pumps on one power supply, two bolts from one bad batch, two sensors calibrated by the same technician: in each case the parallel formula overstates the truth, often badly. When you use $1 - \prod(1 - p_i)$, say what makes the independence plausible.

4.7 Three errors

1. **Treating disjoint as independent.** They are opposites.
2. **Assuming independence because no connection comes to mind.** Absence of an imagined mechanism is not evidence of independence; shared causes are usually invisible from the data alone.
3. **The gambler's fallacy.** For genuinely independent trials, past outcomes change nothing. A machine that has produced nine good bolts is no more "due" a defective one.

4.8 Summary

Week 4 in six lines

- Independence means $P(A \cap B) = P(A)P(B)$, equivalently $P(A \mid B) = P(A)$.
- Disjoint events with positive probability are dependent, never independent.
- Independence survives complementation, which is why "none of them" multiplies.
- Pairwise independence does not give mutual independence.
- Conditional independence and independence are separate properties.
- Series multiplies, parallel multiplies the failures, and both assume what they assume.

Exercises

1. $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cup B) = 0.7$. Are A and B independent? Show your working.
2. Prove that if A and B are independent, so are A^c and B^c.
3. Give an example of two events that are dependent although neither causes the other.

4. Three pumps fail independently, each with probability 0.1 in a given month. Find the probability that: all three fail; none fails; at least one fails.

5. A die is rolled twice. Let A be "first roll is a six", B "second roll is a six" and C "the sum is seven". Check each pair for independence, and then check mutual independence.

6. Show that if A and B are independent and $A \subseteq B$ with $P(A) > 0$, then $P(B) = 1$.

7. Four components each work with probability 0.85, independently. Find the reliability if they are all in series, and if they are all in parallel.

8. A system works if at least two of three independent components work, each working with probability 0.9. Find the reliability.

9. In Example 5, find $P(D^c \mid \text{works})$ and $P(\text{both } C \text{ and } D \text{ work} \mid \text{works})$.

10. Twelve percent of items are defective, independently. Find the probability that a sample of six contains no defective item, and that it contains at least two.

11. A technician argues that two sensors in parallel give 99% reliability because each is 90% reliable. State the assumption, and describe a realistic situation in which the true figure is much lower.

12. Give an example of three events that are mutually independent, and verify all four conditions.

Answers to the odd-numbered exercises

1. From the addition rule, $P(A \cap B) = 0.4 + 0.5 - 0.7 = 0.2$, and $P(A)P(B) = 0.20$. They are equal, so the events are independent.

3. For instance, among students, owning a laptop and living off campus: both are consequences of family income, and neither causes the other.

5. $P(A) = P(B) = 1/6$ and $P(C) = 6/36 = 1/6$. Then $P(A \cap B) = 1/36 = P(A)P(B)$; $P(A \cap C) = P(\{(6,1)\}) = 1/36 = P(A)P(C)$; similarly for B and C. All three pairs are independent. But $P(A \cap B \cap C) = 0$, since two sixes sum to twelve, while $P(A)P(B)P(C) = 1/216$, so they are not mutually independent.

7. Series: $0.85^4 = 0.522$. Parallel: $1 - 0.15^4 = 0.9995$.

9. $P(D^c \mid \text{works}) = 0.1296/0.7776 = 0.1667$ by symmetry with C. For both working, $0.81 \times 0.64/0.7776 = 0.5184/0.7776 = 0.667$.

11. The assumption is that the two sensors fail independently. If both are powered from the same supply, or were calibrated together, or sit in the same overheating cabinet, a single cause takes out both, and the system reliability falls towards that of one sensor, 90%, rather than 99%.

Before next week

Attempt exercises 1 to 8. Week 5 introduces random variables: instead of asking whether an event happened, we attach a number to each outcome and study how that number behaves.